

SACHET KORADA

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EDUCATION

Bachelor of Science in Robotics, Minor in Business Administration
Carnegie Mellon University

2024 – 2028

Relevant Coursework: Principles of Imperative Computation, Computer Systems, Computer Vision, General Robotics, Feedback Control Systems, Sensing & Sensors, Foundational Mathematics of Robotics, Multi-Robot Planning & Coordination

SKILLS

Languages	C++, Python, CUDA, Java, MATLAB
Libraries & Frameworks	OpenCV, NumPy, YOLOv8, ROS 2, Open3D, CUDA, PyTorch
Sensors & Robotics Hardware	LiDAR, RGB-D Cameras, ESP-32, Raspberry Pi, NVIDIA Jetson Orin

EXPERIENCE

Software Engineering Intern

June – August 2025

Hughes Networking System

Gaithersburg, MD

- Enabled large-scale satellite network subsystem stress testing by developing **C++** simulators using **socket programming** and **Google Protobuf**, modeling **100+ concurrent users** interacting with the Anchor Control Processor.
- Improved system reliability by identifying **race conditions** through **multi-threaded communication pipelines** implemented in **C++** and **Linux**, increasing simulation realism for **distributed network testing**.
- Reduced manual testing effort by **automating validation workflows** using **gTest** and **gMock**, creating **20+ unit and integration tests** for **concurrent network components**.

Software Engineering Intern

June – August 2023

Mantaro Product Development Services

Germantown, MD

- Deployed an **open-source ROS 2 autonomy stack** in **C++** and **Python** for an **unmanned ground vehicle**, achieving a stable **50 Hz control loop** for **motors, cameras, and encoders** on **Raspberry Pi** embedded hardware.
- Engineered a modular **UGV** platform architecture using embedded **Linux** and open-source robotics software, lowering system cost by **76% (\$1200 → \$290)** while maintaining **4+ hours of operational runtime**.
- Optimized system initialization pipelines for **ROS nodes** and **embedded software components**, achieving **sub-2 second startup times** for the full robotics control stack.

PROJECTS

CMU RoboBuggy — Multi-Sensor Perception System

August 2026 – Present

- Executed **LiDAR-camera sensor fusion** by projecting 3D point clouds and associating camera detections with **LiDAR clusters** for **obstacle classification** for an **autonomous racing vehicle** operating in real-world conditions.
- Architected a **LiDAR perception pipeline** using **NumPy** and **Open3D** to perform **RANSAC** ground segmentation and obstacle extraction for an autonomous racing vehicle.
- Analyzed perception outputs by constructing **visualization and debugging tools** to **validate autonomous system performance** across recorded driving datasets and live field tests.

Tree of Heaven Detection System

August 2026 – Present

- Applied **photogrammetry techniques** to align **aerial imagery** with **LiDAR point clouds**, fusing **YOLOv8 segmentation** with **3D spatial data** to generate **geospatial heatmaps** of likely Tree-of-Heaven plant locations.
- Trained **YOLOv8 segmentation models** on **1,000+ labeled images**, achieving **0.71 mAP** and enabling automated detection of invasive Tree-of-Heaven plants in field imagery.
- Accelerated large-scale inference by **optimizing GPU pre-processing** and batch inference pipelines using **CUDA-enabled PyTorch**.

Autonomous Search and Rescue Robot

March – April 2025

- Designed and built a **Raspberry Pi** based **unmanned ground vehicle** for navigating **simulated disaster environments** by architecting a **modular robotics software stack** using embedded Linux and **ROS-inspired publisher/subscriber** messaging patterns.
- Implemented **real-time teleoperation** and **video streaming** by developing **socket-based communication** pipelines using **Python** and **OpenCV**, enabling **live control and monitoring** of the robot.